Computer Science & IT



Data Structure & Programming

Array

Lecture No. 03



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Recap of Previous Lecture



Topic

Topic

Topic

Topic

Topic

2-D array Column major order

Lower triangular Matrix RMO

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

LTM :- Column major order

3-D Array

Complexity

UTM

Question : Lower Triangular Matrix

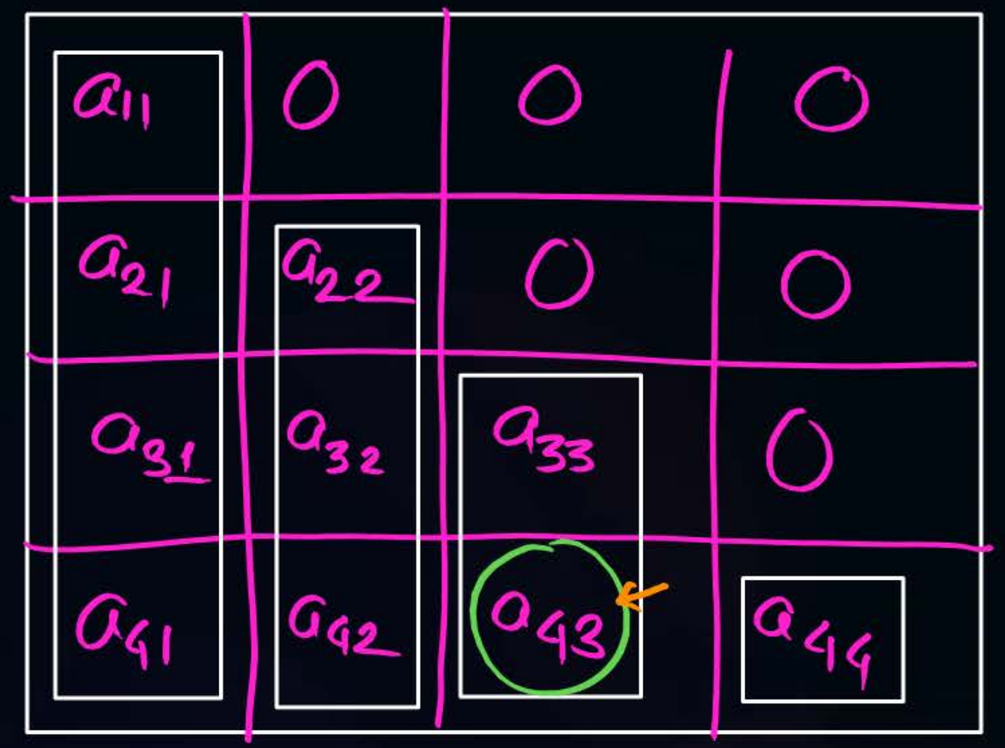
a_{11}	0	0	0
a_{21}	a_{22}	0	0
a_{31}	a_{32}	a_{33}	0
a_{41}	a_{42}	a_{43}	a_{44}

$$BA + \left[\frac{i(i-1)}{2} + j-1\right] \times \text{Size}$$

RMO.

Slide

Question : Lower Triangular Matrix



a_{43} ← 4th Raw, 3rd column

3rd column . No. of colums completed :- 2

No. of elemets :- 4 + 3 = 7 ✓

(4 : first column, 3 : second)

in jth column . No. of element arranged : 1

total : 7 + 1 = 8

Add a_{43} : 100 + 8 × 2 = 116

a_{11}	a_{21}	a_{31}	a_{41}	a_{22}	a_{32}	a_{42}	a_{33}	a_{43}	a_{44}

Slide

Question : Lower Triangular Matrix

4×4

a_{11}	0	0	0
a_{21}	a_{22}	0	0
a_{31}	a_{32}	a_{33}	0
a_{41}	a_{42}	a_{43}	a_{44}

$n \times n$

a_{ij} :- i^{th} Row, j^{th} colum

No. of columns completed :- $j-1$

1st Column :⟶ $n - (1-1)$

2nd column :⟶ $n-1$ $(2-1)$

3rd column ⟶ $n-2$ $3-1$

$j-1^{th}$ Column ⟶ $n-(j-1-1)$

$= n-(j-2)$

a_{11}	a_{21}	a_{31}	a_{41}	a_{22}	a_{32}	a_{42}	a_{33}	a_{43}	a_{44}

Slide

Question : Lower Triangular Matrix

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a_{11}	a_{21}	a_{31}	a_{41}	a_{22}	a_{32}	a_{42}	a_{33}	a_{43}	a_{44}

Slide

No. of element arranged

terms? $j-1$

$$n + n-1 + n-2 + \cdots + n-(j-2)$$

$$n \times (j-1) - \left(1+2+3 \cdots + (j-2)\right) \qquad \frac{n(n+1)}{2}$$

$n : j-2$

$$\boxed{n \times (j-1) - \frac{(j-1)(j-2)}{2}} \qquad \frac{(j-2)(j-1)}{2}$$

In j^{th} column No. of element arranged is $i-j$

$$BA + \left[n(j-1) - \frac{(j-1)(j-2)}{2} + i - j \right] \times Size$$



Topic : Question

100x100

Q A[1...100][1...100] is lower triangular matrix. Base address is 1000 and each element occupies 2-Bytes of space. What is the address of A[50][50] in ~~row~~ column major order ?

n×n

$$BA + \left[n(j-1) - \frac{(j-1)(j-2)}{2} + (i-j) \right] \times Size$$

$$1000 + \left[100 \times 49 - \frac{49 \times 48}{2} + 0 \right] \times 2 = 1000 + [4900 - 1176] \times 2$$

$$1000 + 7448 = 8448$$

Slide

Topic : Lower Triangular Matrix: Column Major Order

$10-(-4)+1 = 15$

$20-6+1 = 15$

Rows = Column

A[-4...10][6...20]

Base Address :- 1000, 2B Size

$BA+\left[\frac{i(i-1)}{2}+(j-1)\right]\times$

A[5][13] ✓

Subtract a No

6 becomes 1

$BA+\left[n(j-1)-\frac{(j-1)(j-2)}{2}+(i-j)\right]\times S_i$

Bound Adjustment

Add a No so that -4 become 1

A[-4..10][6..20]

+5 ((-5 ((

A[1....15][1...15]

A[5][13] J>i

+5 (-5 (

A[10][8]

Slide



Topic : Lower Triangular Matrix: Column Major Order

$A[1..15][1...15]$

RMO

$A[10][8] = 1000 + \left[\frac{10\times 9}{2} + 7\right] \times 2$

1104

2 107
2
86×2
172

CMO

$= 1000 + \left[15\times 7 - \frac{7\times 6}{2} + 2\right] \times 2 = 1172$

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Topic : Lower Triangular Matrix: Column Major Order

Consider a LTM in RMO A[1..15][1...15] stored in memory only non zero element are stored.

if we store Natural No from A[1][1] the the No. stored in A[12][4] is _____

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Topic : Lower Triangular Matrix: Column Major Order

5X5

1				
2	3			
4	5	6		
7	8	9	10	
11	12	13	14	15

$a_{54} \rightarrow \frac{n(n+1)}{2}$

$\frac{4 \times 5}{2} = 10$

12th

$\frac{11 \times 12}{2} = 66$

$a_{54} = BA + \left(\frac{(i(i-1)}{2} + (j-1) \right) \times Size$

$1 + \left(\frac{5 \times 4}{2} + 3 \right)$

$1 + 13 = 14$

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Topic : Lower Triangular Matrix: Column Major Order

$$1 + \left[\frac{12 \times 11}{2} + 4 - 1\right] \times 1 = 1 + 69 = 70$$

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Question : 2-D Array

#Q Suppose you are given with 2-D Array A, where array is arranged in row major order and each row consists of 4 elements . Let the base address is 1000 size of each element is 4B. An element A[i][j] is stored in 1068 then what is value of i+j 7 .
(Lower bound is 1)

Array : 1000 — 1067

68 Bytes

A[5][2]



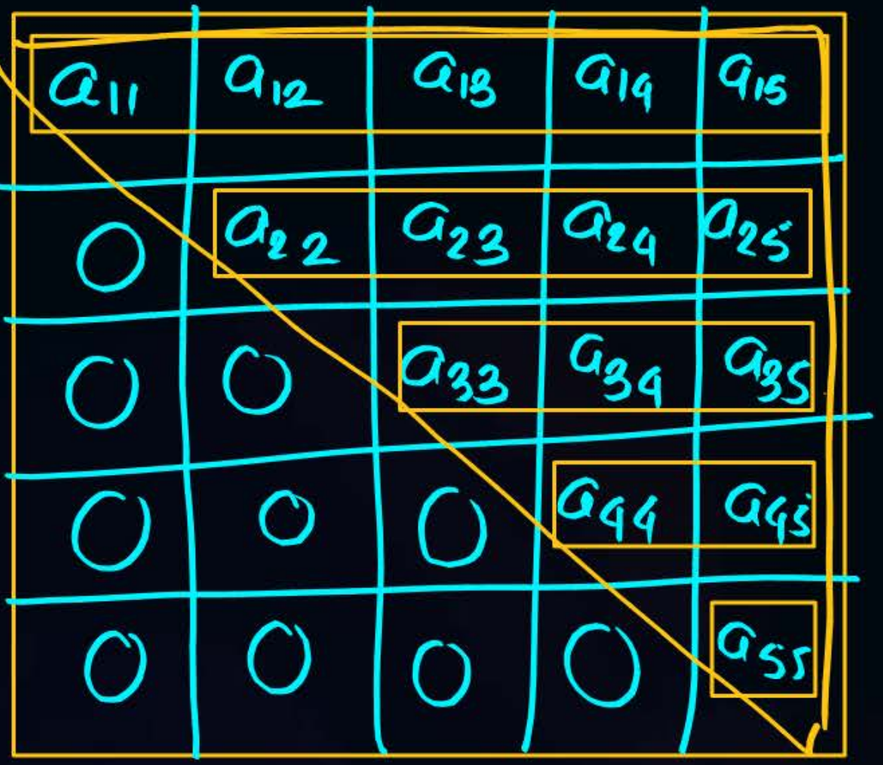
No. of elements = 68/4 = 17

BA + [(i-1)×4+(j-1)] ×4

1000 + [16+1] ×4 = 1068

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Topic : Upper Triangular Matrix



UTM RMO

Similar analysis LTM - CMO

a_{ij}

$i - (i-1)$ Rows

$n + n-1 + n-2 \cdots + (n-(i-2))$

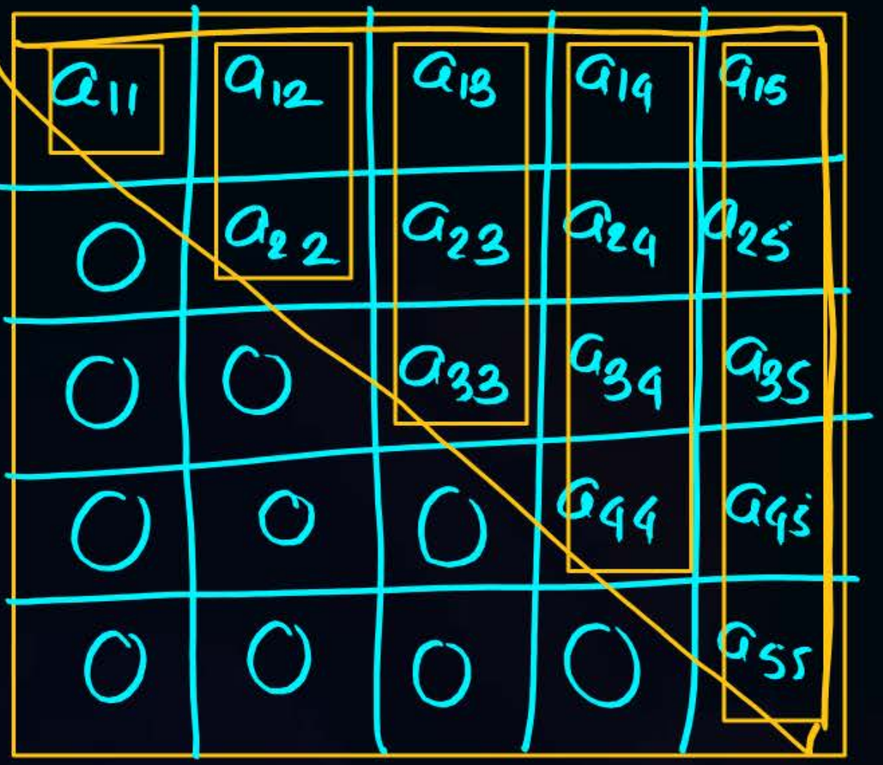
$n(i-1) - \frac{(i-1)(i-2)}{2}$

on ith Row : $j - i$ elements are arranged

$$BA + \left[n(i-1) - \frac{(i-1)(i-2)}{2} + (j-i) \right] \times Size$$

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Topic : Upper Triangular Matrix



UTM CMO

a[i,j], j^{th} column, j-1 columns are completed

$$1+2+3+\cdots+(j-1)=\frac{j(j-1)}{2}$$

on j^{th} column i-1

$$BA+\left[\frac{j(j-1)}{2}+i-1\right]\times Size$$

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2 mins Summary

Topic	CMO - ITM
Topic	Bound Adjustment
Topic	ITM, RMO, CMO
Topic	
Topic	

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THANK - YOU